



**Stemronic AI**

INTELLIGENT SOLUTIONS · INDUSTRIAL IMPACT

# Build Status & Session Context

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Everything shipped so far on the Lead Generation Platform, the key decisions behind it, and what's still open.

**Product** AI B2B lead generation for GCC industrial automation

**Status** Live on Render · persistent storage · single-container deploy

**Stack** FastAPI + SQLite · React/Vite/Tailwind · Groq LLM

**Repo** [github.com/JawadAhmed9/lead-gen](https://github.com/JawadAhmed9/lead-gen) (private)

SECTION 01

# Executive snapshot

Stemronic AI is a live, deployed platform that automatically discovers, enriches, scores, and works B2B leads for industrial-automation sales across the Gulf. It pairs an AI acquisition pipeline with a full sales-operations app (assignment, calling, RFQ pipeline, leaderboards) and an embedded AI Copilot that can search the data and take actions on approval.

| AREA                                     | STATE                                                                    |
|------------------------------------------|--------------------------------------------------------------------------|
| Lead pipeline (collect → enrich → score) | <div>Live</div> Apollo + social sources, AI scoring                      |
| Sales-operations layer                   | <div>Live</div> RBAC, assignment, Call Console, RFQ, leaderboard         |
| AI Copilot (L1–L4)                       | <div>Built</div> ask · act-on-approval · orchestrate · background agents |
| Deployment & persistence                 | <div>Live</div> Render Starter + 1 GB disk, data survives redeloys       |
| Branding & mobile                        | <div>Live</div> Stemronic rebrand, responsive, split-screen login        |
| Email delivery (Brevo)                   | <div>Pending</div> code ready; awaiting domain DNS + keys                |
| WhatsApp outreach                        | <div>Scaffold</div> built; connect a Meta number to activate             |
| Phase-3 (call, tender, IoT bridge)       | <div>Planned</div> blueprints on the Roadmap page                        |

SECTION 02

# What's been built

## Lead acquisition pipeline

- Apollo two-step sourcing (search → bulk-match) with a filter-rotation engine that sweeps every location × company-size segment so runs never re-pull the same companies.
- Enrichment — Hunter.io email verification + Wappalyzer tech-stack detection.
- AI ICP scoring — Groq `llama-3.3-70b-versatile`, 0–100 score with reason; gate at `MIN ICP SCORE = 45` flips leads to *queued* (sales-ready).
- Free social sources — Reddit + industrial forums (PLCtalk, Eng-Tips) collectors wired as a "Collect Social" pipeline step; AI signal-extractor moved from Anthropic to Groq so it needs no extra key.
- Excel import — reads every sheet; a real customer database of 8,573 companies was cleaned and imported (~13,500+ leads in system).

## Sales-operations layer

- RBAC + teams — roles admin / manager / agent / viewer; managers own agents; permissions enforced server-side.
- Assignment — single & bulk, with re-assignment guards; agents only see their own leads.
- Guided Call Console — timer, category talk-tracks, live notes, 1–10 interest rating, RFQ capture, follow-up scheduling.
- Editable call scripts by client category — admins, managers *and agents* can edit; 8 seeded scripts incl. 5 industry verticals (F&B, Pharma, Logistics, Automotive, Utilities).
- Tasks & cadences — follow-ups + multi-touch sequences (Standard / Fast / Nurture) that auto-schedule the next step.

## RFQ pipeline & performance

- RFQ pipeline — drag-and-drop Kanban (New → Quoted → Won → Lost) with a weighted forecast (stage weights 0.3 / 0.6 / 1.0). Every RFQ secured on a call auto-lands here.
- Leaderboard — RFQ-centric weighted points: `RFQ 40` `reply 15` `call 5` `email 3` (+ per-\$1k RFQ value). Podium + sortable table.
- Targets & charts — managers set daily call/RFQ targets; target-vs-actual graphs on the agent dashboard and leaderboard. (Deals removed; measured on RFQs.)

## AI Copilot — the "Ask & Act" agent (L1–L4)

A futuristic corner orb (⌘J) that expands into a glassy panel, mounted on every screen. Built in layers, all read-safe and RBAC-scoped:

| LAYER            | CAPABILITY                                                                                                                                                           |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L1 - Ask         | Read-only: search leads by spec, answer pipeline questions, navigate. Structured tools + a guarded read-only SQL tool (SELECT-only, sandboxed) for ad-hoc analytics. |
| L2 - Act         | Propose-only write actions — draft email, assign, create RFQ — each rendered as an approval card; execution hits the real endpoint (re-checks permissions).          |
| L3 - Orchestrate | Multi-step in one ask ("draft emails to my top 3 leads" → 3 approval cards) via the tool loop.                                                                       |
| L4 - Background  | Prospecting + stale-RFQ agents that drop proposals into an approval inbox (badge + Run-now). Idempotent, scoped.                                                     |

Engine: Groq tool-calling loop over the app's own REST API; every question is audit-logged; lead text is treated as data, never instructions. Demo panel pre-seeds live Q&A + sample query chips.

### Channels, notifications & roadmap

- **In-app notifications** — top-bar bell with badge: overdue/today tasks + Copilot inbox items, click-through.
- **WhatsApp outreach (scaffold)** — Compose channel toggle (Email / WhatsApp); Meta Cloud API send, gated until a business number is connected.
- **Channels & Roadmap page** — Phase-3 blueprint cards: click-to-call (Twilio), tender/RFP monitoring, post-sale/IoT bridge.

### Branding, UX & deployment

- **Stemronic rebrand** — hexagon logo (crisp SVG), navy + electric-blue theme, split-screen login with glowing hero, favicon, full accent pass; demo credentials removed.
- **Mobile responsive** — sidebar → drawer with hamburger, stacking grids, scrollable tables, full-width panels.
- **Deployment** — multi-stage Docker on Render (Starter plan) with a 1 GB persistent disk at `/var/data`; DB seeds once from the baked-in baseline, then persists. SQLite in `WAL` mode for concurrent agents. User & script seeding is idempotent (survives resets).

SECTION 03

Key configuration & decisions

| SETTING               | VALUE / DECISION                                                                                                                                                                              |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scoring & Copilot LLM | Groq <code>llama-3.3-70b-versatile</code> (free). Recommended paid upgrade: OpenAI <code>gpt-5.6-luna</code> for both (fast/cheap), <code>gpt-5.6-terra</code> for heavier Copilot reasoning. |
| ICP threshold         | <code>MIN_ICP_SCORE = 45</code> (queue), <code>HIGH_INTENT_SCORE = 70</code>                                                                                                                  |
| Leaderboard weights   | RFQ 40 · reply 15 · call 5 · email 3 · +1 per \$1k RFQ value                                                                                                                                  |
| RFQ stage weights     | new 0.3 · quoted 0.6 · won 1.0 · lost 0.0                                                                                                                                                     |
| Seeded logins         | 3 Stemronic admins (jawad / khan / mohsin @stemronic.com, pw <code>stemronic432</code> ) + demo accounts (kept per your call)                                                                 |
| Data persistence      | <code>DATA_DIR=/var/data</code> on Render disk; SQLite WAL + 30s busy-timeout                                                                                                                 |
| Target market         | 6 GCC countries · manufacturing, oil&gas, pharma · 50–5,000 employees                                                                                                                         |
| Security              | Salted SHA-256 passwords, bearer-token sessions, secrets in env only ( <code>.env</code> git/docker-ignored), full audit log                                                                  |

NOTABLE ENGINEERING FIXES THIS SESSION

Fixed the persistence gap (ephemeral → disk), made user/script seeding idempotent so colleagues' logins appear on the live server, moved the social signal-extractor to Groq, and resolved a GitHub push-protection block caused by a stray duplicate copy ( `Lead-Gen-Platform/` ) that carried a real key — now untracked and git-ignored.

SECTION 04

Integrations & keys

| SERVICE               | PURPOSE                                                | STATUS                                                     |
|-----------------------|--------------------------------------------------------|------------------------------------------------------------|
| Groq                  | AI scoring, email drafting, signal extraction, Copilot | Active (free)                                              |
| Apollo                | Primary lead sourcing                                  | Active                                                     |
| Hunter / Wappalyzer   | Email verify + tech stack                              | Active                                                     |
| Brevo                 | Email delivery (HTTPS API)                             | Pending DNS auth at GoDaddy + set BREVO * in Render        |
| Reddit API            | Social intent leads                                    | Optional add REDDIT_CLIENT_ID/SECRET (forums work without) |
| WhatsApp (Meta Cloud) | WhatsApp outreach                                      | Not connected add WHATSAPP_TOKEN/PHONE_ID                  |
| OpenAI                | Optional premium LLM + voice                           | Not wired toggle recommended                               |

## SECTION 05

## Open items & next steps

### ACTION NEEDED FROM YOU / IT

- Brevo email go-live — apply the DNS records (Brevo code + DKIM×2 + branding CNAMEs) at GoDaddy; DMARC already ✓. Then set `BREVO_SENDER_EMAIL` in Render. (A ready-to-send DNS request PDF was produced.)
- Reddit keys — create a "script" app at reddit.com/prefs/apps to switch on Reddit social sourcing.
- WhatsApp — connect a Meta WhatsApp Business number to activate sending.

### BUILD BACKLOG (PROPOSED / DISCUSSED)

- OpenAI provider toggle — one env switch (Groq ↔ OpenAI `gpt-5.6-luna`) across scorer + Copilot + email-gen.
- Multilingual Copilot — Italian/Arabic auto-detect (prompt change; near-free).
- Voice Copilot — STT (`gpt-4o-transcribe`) → agent → TTS (`gpt-4o-mini-tts`), or live `gpt-realtime`.
- Background-agent schedule — Render Cron hitting `/api/agent/jobs/run` for nightly prospecting.
- Phase-3 — click-to-call, tender monitoring, post-sale/lot bridge (blueprints on Roadmap).
- Strategic ideas — AI quote/BOQ generator, self-learning ICP from won/lost, white-label multi-tenant.

## SECTION 06

## Session context — how we got here

A chronological summary of this working session, for continuity.

1. Finished *Wave 2*: editable call scripts by category + the RFQ Pipeline Kanban; fixed an RFQ auto-create bug.
2. Made the app *production-persistent* — moved SQLite onto a Render disk, idempotent seeding so real logins survive; deployed live.
3. Fixed live bugs: routing for Scripts/RFQ pages, a Leaderboard crash, and a broken Tailwind config (white sidebar); removed demo creds from login.
4. Rebranded to *Stemronic AI* (logo, theme, split-screen login) and made the whole app mobile-responsive.
5. Produced documents: technical architecture PDF, C-level sales deck, admin + agent user manuals, a DNS setup request for the boss, and blueprint diagrams.
6. Worked the *Brevo* email domain-authentication path (still pending GoDaddy DNS).
7. Added *free social sourcing* (Reddit + forums) as a pipeline step; enabled agent script editing; added 5 industry scripts.
8. Designed and built the AI Copilot L1→L4 (plan → UI preview → engine → approval writes → background inbox), all tested.
9. Added *notifications*, a WhatsApp scaffold, and the *Channels & Roadmap* page for Phase-3 items.
10. Resolved a GitHub secret-scanning push block (stray duplicate folder), then explored future directions — Italian + voice, and the best paid OpenAI models (recommended GPT-4.5 Turbo).

### GUIDING PRINCIPLES HELD THROUGHOUT

Everything real (no demo hardcoding), the pipeline never broken (additive changes), secrets kept out of git, the repo private, and every new capability verified before shipping.